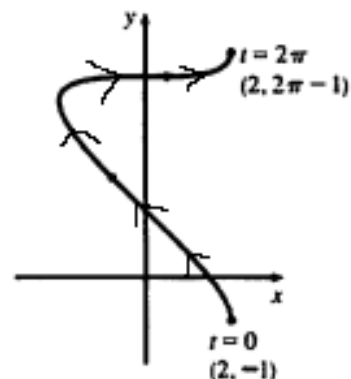


5.1 Parametric Worksheet

1. Sketch the curve of the following parametric function by plotting points. Indicate with arrows the direction in which the curve is traced as t increases.

$$x = 2 \cos(t), \quad y = t - \cos(t), \quad 0 \leq t \leq 2\pi$$

t	0	$\pi/2$	π	$3\pi/2$	2π
x	2	0	-2	0	2
y	-1	$\pi/2$	$\pi+1$	$3\pi/2$	$2\pi-1$

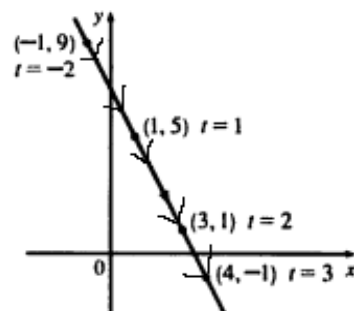


2. Consider the parametric curve given by the parametric equations

$$x = 1 + t, \quad y = 5 - 2t, \quad -2 \leq t \leq 3$$

- (a) Sketch the curve of the following parametric function by plotting points. Indicate with arrows the direction in which the curve is traced as t increases.

t	-2	1	0	1	2	3
x	-1	0	1	2	3	4
y	9	7	5	3	1	-1



- (b) Find a Cartesian equation for this curve.

We note that if $x = 1 + t$ then $t = x - 1$ which implies that

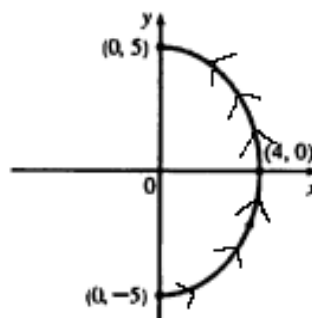
$$y = 5 - 2(x - 1) \Leftrightarrow y = -2x + 7 \text{ where } -1 \leq x \leq 4$$

3. Consider the parametric curve given by the parametric equations

$$x = 4 \cos(\theta), \quad y = 5 \sin(\theta), \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

- (a) Sketch the curve of the following parametric function by plotting points. Indicate with arrows the direction in which the curve is traced as t increases.

t	$-\pi/2$	$-\pi/4$	0	$\pi/4$	$\pi/2$
x	0	$2\sqrt{2}$	4	$2\sqrt{2}$	0
y	-5	$\frac{-5\sqrt{2}}{2}$	0	$\frac{5\sqrt{2}}{2}$	5



- (b) Find a Cartesian equation for this curve.

We note that if $x = 4 \cos(\theta)$ & $y = 5 \sin(\theta)$ then $\frac{x^2}{16} = \cos^2(\theta)$ & $\frac{y^2}{25} = \sin^2(\theta)$ which implies that $\frac{x^2}{16} + \frac{y^2}{25} = \cos^2(\theta) + \sin^2(\theta) \Leftrightarrow \frac{x^2}{16} + \frac{y^2}{25} = 1$ where $0 \leq x \leq 4$ & $-5 \leq y \leq 5$.

(or we could proceed in the following way)

We note that if $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ then we can say that $y = 5 \sin(\theta)$ implies that $\theta = \sin^{-1}\left(\frac{y}{5}\right)$.

Therefore we can conclude that $x = 4 \cos\left(\sin^{-1}\left(\frac{y}{5}\right)\right)$

4. Consider the curve given by the parametric equations

$$x = 2e^{3t} + t^2, \quad y = 5e^{2t}, \quad 0 \leq t \leq 1$$

- (a) Find a formula for $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{10e^{2t}}{6e^{3t} + 2t}$$

- (b) What is the equation of the line tangent to the curve at the point $(2, 5)$?

We note that the point $(2, 5)$ is reached when $5 = 5e^{2t}$ which occurs when $t = 0$. We can also verify that when $t = 0$, $x = 2$. Now we have that the slope of the tangent line is

$$\left. \frac{dy}{dx} \right|_{t=0} = \frac{10e^{2(0)}}{6e^{3(0)} + 2(0)} = \frac{10}{6} = \frac{5}{3}.$$

So the equation of the tangent line is given by:

$$y - 5 = \frac{5}{3}(x - 2) \Leftrightarrow y = \frac{5}{3}x + \frac{5}{3}$$

(c) Set up, but do not evaluate, an integral with respect to t that represent the arc length of the curve.

We must first check the necessary conditions. We note (from the work in part (b)) that both x' and y' are continuous functions on $0 \leq t \leq 1$ and clearly since $y = 5e^{2t}$ is a strictly increasing function on $0 \leq t \leq 1$ the given parametric curve is only traced once on this interval. Therefore the arc length of the curve is given by:

$$L = \int_0^1 \sqrt{(6e^{3t} + 2t)^2 + (10e^{2t})^2} dt$$

5. Consider the curve given by the parametric equations

$$x = 3 \cos\left(\frac{t}{2}\right), \quad y = t^2, \quad 0 \leq t \leq 4\pi$$

(a) What are the slopes of the tangent lines to the curve at the point when $x = 0$.

We note that $x = 0$ when $0 = 3 \cos\left(\frac{t}{2}\right)$. Based on the restrictions for t we can see that it must be the case that $t = \pi$ or $t = 3\pi$. Therefore we are interested in the points $(0, \pi^2)$ & $(0, 9\pi^2)$.

Note that $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2t}{-\frac{3}{2} \sin\left(\frac{t}{2}\right)} = \frac{-4t}{3 \sin\left(\frac{t}{2}\right)}$ and so the slopes of the desired tangent lines are:

$$\left. \frac{dy}{dx} \right|_{t=\pi} = \frac{-4\pi}{3 \sin\left(\frac{\pi}{2}\right)} = -\frac{4\pi}{3} \quad \text{and} \quad \left. \frac{dy}{dx} \right|_{t=3\pi} = \frac{-12\pi}{3 \sin\left(\frac{3\pi}{2}\right)} = \frac{12\pi}{3} = 4\pi.$$

So the equations of the tangent lines are:

$$y - \pi^2 = \frac{-4\pi}{3}(x - 0) \Leftrightarrow y = \frac{-4\pi}{3}x + \pi^2 \quad \& \quad y - 9\pi^2 = 4\pi(x - 0) \Leftrightarrow y = 4\pi x + 9\pi^2$$

(b) Set up, but do not evaluate, an integral with respect to t that represents the arc length of the curve.

We must first check the necessary conditions. We note (from the work in part (a)) that both x' and y' are continuous functions on $0 \leq t \leq 4\pi$ and clearly since $y = t^2$ is a strictly increasing function on $0 \leq t \leq 4\pi$ the given parametric curve is only traced once on this interval. Therefore, the arc length of the curve is given by:

$$L = \int_0^{4\pi} \sqrt{\left(-\frac{3}{2} \sin\left(\frac{t}{2}\right)\right)^2 + (2t)^2} dt$$

(c) Determine the second derivative of the parametric curve give by $x = 3\cos\left(\frac{t}{2}\right)$, $y = t^2$, $0 \leq t \leq 4\pi$.

We saw in part (a) that $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2t}{-\frac{3}{2}\sin\left(\frac{t}{2}\right)} = \frac{-4t}{3\sin\left(\frac{t}{2}\right)}$.

Now we calculate $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt} = \frac{\left(\frac{-\frac{3}{2}\sin\left(\frac{t}{2}\right) \cdot 2 - 2t\left(-\frac{3}{4}\cos\left(\frac{t}{2}\right)\right)}{\frac{9}{4}\sin^2\left(\frac{t}{2}\right)}\right)}{-\frac{3}{2}\sin\left(\frac{t}{2}\right)} = \frac{-3\sin\left(\frac{t}{2}\right) + \frac{3}{2}t\cos\left(\frac{t}{2}\right)}{-\frac{27}{8}\sin^3\left(\frac{t}{2}\right)}$